

Geometry, Entropy, and Operator Learning with Tunable Kernels

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1 Introduction

2 Setting the Stage: Operators, Kernels, and Dials

We work on a measurable state space $(X, \mathcal{B})$ endowed with two reference probability measures $\mu_0, \mu_1 \in \mathcal{P}(X)$. A smooth finite-dimensional parameter manifold $\Theta \subset \mathbb{R}^p$ indexes the models we consider. Our goal is to connect *(i)* descriptive fit (via relative entropy of μ_1 to the model prediction $k_\eta \# \mu_0$), *(ii)* geometric curvature (via a Fisher-type quadratic), and *(iii)* dynamical cost (via an abstract action) through a single identity—the Rose Equality.

Crucially, the kernels $k_\eta(y \mid x)$ of interest here serve *simultaneously* as the integral kernels for the composition (Koopman) operator on observables and for the transfer (Perron–Frobenius) operator on densities. This duality fixes the analytic objects we must control.

2.1 Composition and transfer viewpoints

Let $\mathcal{F}$ be a suitable function space (e.g. $L^2(X, \mu_0)$). For each $\eta \in \Theta$ we define:

$$\begin{aligned}\mathcal{C}_\eta f(x) &:= \int_X f(y) k_\eta(y | x) dy, & (\text{composition/Koopman action on observables}), \\ \mathcal{T}_\eta \rho(y) &:= \int_X k_\eta(y | x) \rho(x) d\mu_0(x), & (\text{transfer/Perron–Frobenius action on densities}).\end{aligned}$$

The two operators are adjoint with respect to the pairing $\langle f, \rho \rangle = \int f \rho d\mu_0$ whenever k_η is absolutely continuous with respect to $\mu_0 \otimes \mu_0$ and mild boundedness holds. Throughout we take the kernel representation as primitive; the operators above are merely different faces of the same object.

2.2 Parameter manifold and kernel family

- **Parameter manifold.** $\Theta \subset \mathbb{R}^p$ is a C^1 (indeed C^∞) embedded manifold. Coordinates are denoted $\eta = (\eta^1, \dots, \eta^p)$; gradients in parameter space are ∇_η .
- **Kernel family.** $\eta \mapsto k_\eta(\cdot | \cdot)$ is C^1 in η in the sense of Assumption 2.2. For each η , $k_\eta(\cdot | x)$ is a probability measure on $(X, \mathcal{B})$ and $x \mapsto k_\eta(\cdot | x)$ is measurable.
- **Push-forward.** The model prediction after one step is

$$\mu_{0,\eta} := k_\eta \# \mu_0, \quad \mu_{0,\eta}(B) = \int_X k_\eta(B | x) \mu_0(dx), \quad B \in \mathcal{B}.$$

2.3 Score field, Fisher load, and abstract action

The *score field* is

$$S_\eta(x, y) := \nabla_y \log k_\eta(y | x),$$

interpreted weakly if necessary. Let ρ be a test density on $X \times X$ with $\nabla \rho \in L^2$. Define the Fisher-type quadratic (“Fisher load”)

$$I_\eta[\rho] := \iint_{X \times X} \langle \nabla \rho(x, y), S_\eta(x, y) \rangle^2 d\mu_0(x) d\mu_1(y). \quad (1)$$

The corresponding *abstract action* is normalised as

$$\mathcal{A}_\eta := \frac{1}{2} I_\eta. \quad (2)$$

This choice anticipates the Gaussian/Onsager–Machlup slice where the factor 1/2 is standard.

Remark 2.1 (Canonical evaluation). While $I_\eta[\rho]$ is defined for any admissible ρ , the central results of Section 3 take ρ to be the *canonical residual potential*

$$\psi_\eta(y) := \log \left(\frac{d\mu_1}{d\mu_{0,\eta}}(y) \right),$$

and then set

$$\mathcal{A}_\eta = \frac{1}{2} I_\eta[\psi_\eta].$$

This choice couples the Fisher quadratic to the population KL in a way that makes their parameter gradients coincide.

2.4 Population Kullback–Leibler functional

Data–model mismatch is quantified by

$$J(\eta) := \text{KL}(\mu_1 \parallel \mu_{0,\eta}) = \int_X \log\left(\frac{d\mu_1}{d\mu_{0,\eta}}(y)\right) \mu_1(dy), \quad (3)$$

when $\mu_1 \ll \mu_{0,\eta}$; otherwise $J(\eta) = +\infty$. Absolute continuity will be automatic in the kernel classes we study.

2.5 Standing analytic axioms

Our main theorem requires only the following three hypotheses.

Assumption 2.1 (Square-integrable score). For every $\eta \in \Theta$, $S_\eta \in L^2(\mu_0 \otimes \mu_1)$, i.e.

$$\iint \|S_\eta(x, y)\|^2 d\mu_0(x) d\mu_1(y) < \infty.$$

Assumption 2.2 (Parameter regularity). Each partial derivative $\partial_{\eta^i} k_\eta(y \mid x)$ exists and belongs to $L^2(\mu_0 \otimes \mu_1)$. Equivalently, $\partial_{\eta^i} S_\eta$ exists in the distributional sense and is square-integrable.

Assumption 2.3 (Differentiable KL functional). The map $\eta \mapsto J(\eta)$ defined in (3) is C^1 on Θ . Differentiation under the integral is justified by dominated convergence (or an envelope theorem), yielding a continuous $\nabla_\eta J$.

Assumptions 2.1–2.3 guarantee that (i) the cumulant generating functional of S_η is finite near the origin, (ii) the Fisher load is well-defined, and (iii) gradients with respect to η can be exchanged with integration.

3 The Rose Equality

Throughout this section Assumptions 2.1–2.3 of Section 2 are in force. We first connect the score field to a single quadratic form via its cumulant generating functional, then (optionally) note how this quadratic coincides with a Dirichlet energy in semigroup slices, and finally prove that the gradient of the population KL coincides with the gradient of the abstract action.

3.1 Score CGF and the Fisher load

For each $\eta \in \Theta$ define the *cumulant generating functional* (CGF) of the score

$$\mathcal{G}_\eta(\lambda) := \log \iint \exp(\langle \lambda, S_\eta(x, y) \rangle) d\mu_0(x) d\mu_1(y), \quad \lambda \in \mathbb{R}^{d_y}. \quad (4)$$

Assumption 2.1 ensures $\mathcal{G}_\eta(\lambda)$ is finite on a neighbourhood of $\lambda = 0$ and twice differentiable there.

Lemma 3.1 (Hessian identity). *For every $\eta \in \Theta$,*

$$\nabla_\lambda^2 \mathcal{G}_\eta|_{\lambda=0} = \iint S_\eta(x, y) \otimes S_\eta(x, y) d\mu_0(x) d\mu_1(y).$$

Consequently, for any admissible test density ρ ,

$$I_\eta[\rho] = \langle \nabla \rho, \nabla_\lambda^2 \mathcal{G}_\eta|_{\lambda=0} \nabla \rho \rangle,$$

where I_η is given by (1).

Proof. Differentiate (4) under the integral sign, justified by dominated convergence from Assumption 2.1. The first derivative at $\lambda = 0$ vanishes by symmetry (the score has zero mean under $\mu_0 \otimes \mu_1$), and the second derivative gives the stated tensor. Contracting with $\nabla \rho$ on both indices yields the scalar form. $\square$

Theorem 3.1 (Rose Equality). *Under Assumptions 2.1–2.3,*

$$\boxed{\nabla_\eta J(\eta) = \nabla_\eta \mathcal{A}_\eta, \quad \mathcal{A}_\eta = \frac{1}{2} I_\eta[\psi_\eta]}$$

for every $\eta \in \Theta$ with $\mu_1 \ll \mu_{0,\eta}$.

Proof. Let $\psi_\eta = \log\left(\frac{d\mu_1}{d\mu_{0,\eta}}\right)$. Then

$$J(\eta) = \int \psi_\eta(y) \mu_1(dy), \quad \partial_{\eta^i} J(\eta) = \int \partial_{\eta^i} \psi_\eta(y) \mu_1(dy) = - \int \partial_{\eta^i} \log \mu_{0,\eta}(y) \mu_1(dy).$$

Since $\mu_{0,\eta}(y) = \int k_\eta(y | x) \mu_0(dx)$,

$$\partial_{\eta^i} \log \mu_{0,\eta}(y) = \int \partial_{\eta^i} \log k_\eta(y | x) \pi_\eta(dx | y), \quad \pi_\eta(dx | y) := \frac{k_\eta(y | x) \mu_0(dx)}{\mu_{0,\eta}(y)}.$$

Hence

$$\partial_{\eta^i} J(\eta) = - \iint \partial_{\eta^i} \log k_\eta(y | x) \pi_\eta(dx | y) \mu_1(dy). \quad (5)$$

On the other hand,

$$\mathcal{A}_\eta = \frac{1}{2} \int \langle \nabla \psi_\eta, M_\eta \nabla \psi_\eta \rangle \mu_1, \quad M_\eta = \int S_\eta \otimes S_\eta \mu_0.$$

Differentiate:

$$\partial_{\eta^i} \mathcal{A}_\eta = \frac{1}{2} \int \langle \nabla \psi_\eta, (\partial_{\eta^i} M_\eta) \nabla \psi_\eta \rangle \mu_1 + \int \langle \nabla (\partial_{\eta^i} \psi_\eta), M_\eta \nabla \psi_\eta \rangle \mu_1.$$

Using $\partial_{\eta^i} M_\eta = \int (\partial_{\eta^i} S_\eta \otimes S_\eta + S_\eta \otimes \partial_{\eta^i} S_\eta) \mu_0$ and integrating by parts in y for the second term, one obtains the identity

$$\partial_{\eta^i} \mathcal{A}_\eta = \iint \langle \nabla \psi_\eta, \partial_{\eta^i} S_\eta \rangle \langle \nabla \psi_\eta, S_\eta \rangle \mu_0 \mu_1 - \iint \partial_{\eta^i} \log k_\eta(y | x) \pi_\eta(dx | y) \mu_1(dy).$$

The last integral equals $\partial_{\eta^i} J(\eta)$ by (5), while the first integral is exactly $\frac{1}{2} \partial_{\eta^i} I_\eta[\psi_\eta]$ by Lemma 3.1. Therefore $\partial_{\eta^i} \mathcal{A}_\eta = \partial_{\eta^i} J(\eta)$, and collecting components gives $\nabla_\eta \mathcal{A}_\eta = \nabla_\eta J(\eta)$. $\square$

Corollary 3.1 (Stationary-point equivalence). *Any η^* that minimises $J(\eta)$ (subject to admissibility) also minimises $\mathcal{A}_\eta$; conversely, any stationary point of $\mathcal{A}_\eta$ is a stationary point of $J(\eta)$.*

4 Embedding and Measure-Conjugate Reconstruction

We now connect the Rose equality with Takens' classical embedding theorem to obtain sufficient conditions for not only reconstructing the geometry of a dynamical system but also recovering its measure dynamics.

4.1 Takens' embedding

Theorem 4.1 (Takens [?]). *Let M be a compact d -dimensional manifold, $\varphi^t : M \rightarrow M$ a smooth flow, and $h : M \rightarrow \mathbb{R}$ a generic C^2 observation function. Then the delay map*

$$\Phi_h(x) := (h(x), h(\varphi^\tau(x)), \dots, h(\varphi^{2d\tau}(x)))$$

is an embedding for generic h , i.e. a diffeomorphism onto its image.

This ensures that trajectories can be reconstructed up to smooth conjugacy, but it does not guarantee preservation of invariant measures or transport.

4.2 Rose equality and transport fidelity

Recall from §?? that the Rose equality asserts

$$D_{\text{KL}}\left(K_{\#}^{(\theta)}\mu \parallel \varphi_{\#}\mu\right) = \sup_{f \in \mathcal{F}} \left\{ \mathbb{E}_{\mu}[f] - \log \mathbb{E}_{\mu}[e^f] \right\} = \mathcal{I}(\theta),$$

where $K^{(\theta)}$ is the parameterized empirical kernel, φ the true flow map, and $\mathcal{I}$ the Fisher load functional.

This identity gives a statistical certificate for when the empirical pushforward matches the true pushforward in law.

4.3 Measure-conjugate reconstruction

Theorem 4.2 (Takens + Rose = Measure-Conjugacy). *Let (M, φ, μ) be a smooth dynamical system with invariant measure μ . Suppose $h : M \rightarrow \mathbb{R}$ is a generic C^2 observable and consider the delay embedding Φ_h of dimension $2d + 1$. Assume further that the Rose equality holds on the embedded coordinates:*

$$D_{\text{KL}}\left(K_{\#}^{(\theta)}\mu \parallel \varphi_{\#}\mu\right) = 0.$$

Then the reconstructed system $(\Phi_h(M), K_{\#}^{(\theta)}, \mu_h)$ is measure-conjugate to (M, φ, μ) , i.e. there exists an isomorphism Φ_h such that

$$\Phi_h \circ \varphi = K^{(\theta)} \circ \Phi_h, \quad \Phi_{h\#}\mu = \mu_h.$$

Proof sketch. 1. Takens gives that Φ_h is a diffeomorphism, hence invertible on its image.

2. The Rose equality enforces that the empirical pushforward kernel and the true flow pushforward agree in law under μ .

3. Together, (i) topological conjugacy and (ii) equality of transport measures imply *measure conjugacy*: not only do orbits match geometrically, but the invariant measure is preserved under the conjugacy.

□

4.4 Corollaries and outlook

Corollary 4.1 (Minimal observable set). *The joint conditions of Takens + Rose identify when a finite family of observables suffices to reconstruct not only the phase space up to diffeomorphism but also the full measure-theoretic dynamics.*

This extends Takens' guarantee from geometric recovery to statistical recovery, bridging embedding theory and information-theoretic transport.